

Skills Summary

Slope and Rate of Change — Inverted and Sign-Reversed Errors

Use this reference while completing your worksheet or when studying.

Skill 1 — Slope from two points, right side up

Formula:

$$m = \frac{\text{rise}}{\text{run}} = \frac{y_2 - y_1}{x_2 - x_1}$$

Two rules that prevent both classic errors:

- **Rise on top.** If your numerator is a difference of x 's, the fraction is *inverted* — your answer is the reciprocal of the slope.
- **Same order top and bottom.** Pick which point is 1 and which is 2, then use that order in both differences. Swapping one of them *reverses the sign*.
- **Predict the sign first:** if y goes down as x goes up, the slope is negative. A sign you predicted rejects a reversed answer instantly.

Example: Find the slope through $(-1, 6)$ and $(3, -2)$.

Step 1. y drops from 6 to -2 while x rises, so the answer must be negative — fix that prediction before computing.

Step 2. Point 1 is $(-1, 6)$, point 2 is $(3, -2)$: $m = \frac{-2 - 6}{3 - (-1)} = \frac{-8}{4}$. The slope is $m = -2$, negative as predicted.

Try it: Find the slope through $(2, -3)$ and $(6, 9)$.

Check: $m = -2$

Skill 2 — Rate of change from a table

Rule: a table's rate of change is a slope — the change in the measured amount over the change in time:

$$\text{rate} = \frac{\text{later amount} - \text{earlier amount}}{\text{later time} - \text{earlier time}}$$

Later minus earlier, in both. Written that way the sign takes care of itself: a falling quantity gives a negative numerator and therefore a negative rate. Never subtract "big minus small" to keep the numbers friendly — that throws away the sign that carries the meaning.

Example: A tank holds 45 litres at hour 0 and 17 litres at hour 7. Find the rate of change.

Step 1. The amount is falling, so the rate must come out negative — keep both subtractions later-minus-earlier so the sign appears on its own.

Step 2. $\frac{17 - 45}{7 - 0} = \frac{-28}{7}$. The rate is -4 litres per hour.

Try it: A tank holds 39 litres at hour 1 and 9 litres at hour 6. Find the rate of change.

check: $9 - 39 \div 5$ litres per hour

Skill 3 — Rate of change in a story

Rule: in a story, the quantity being measured goes on top and the time goes underneath — ask “how much per one unit of time?” and the units name the fraction for you (cm per minute = $\text{cm} \div \text{minutes}$).

Two checks that catch an inverted answer:

- Read the units of your fraction back. Minutes per centimetre is not what the question asked.
- Multiply your rate by the elapsed time. It must reproduce the total change.

Example: A pool’s depth is 12 cm at minute 2 and 3 cm at minute 8. Find the rate of change of the depth.

Step 1. The question asks for cm per minute, so depth goes on top and time underneath — that decision alone prevents the inverted answer.

Step 2. $\frac{3 - 12}{8 - 2} = \frac{-9}{6}$, and $6 \times 1.5 = 9$ cm lost in those 6 minutes ✓ The rate is **−1.5** cm per minute.

Try it: A mug’s temperature is 15 degrees at minute 1 and 3 degrees at minute 5. Find the rate of change.

check: $3 - 15 \div 4$ degrees per minute

(!) Watch out: An inverted slope and a sign-reversed slope both keep the *digits* you computed, so re-checking the arithmetic never finds them. Check the *set-up* instead: what is on top, and did both differences run the same way?